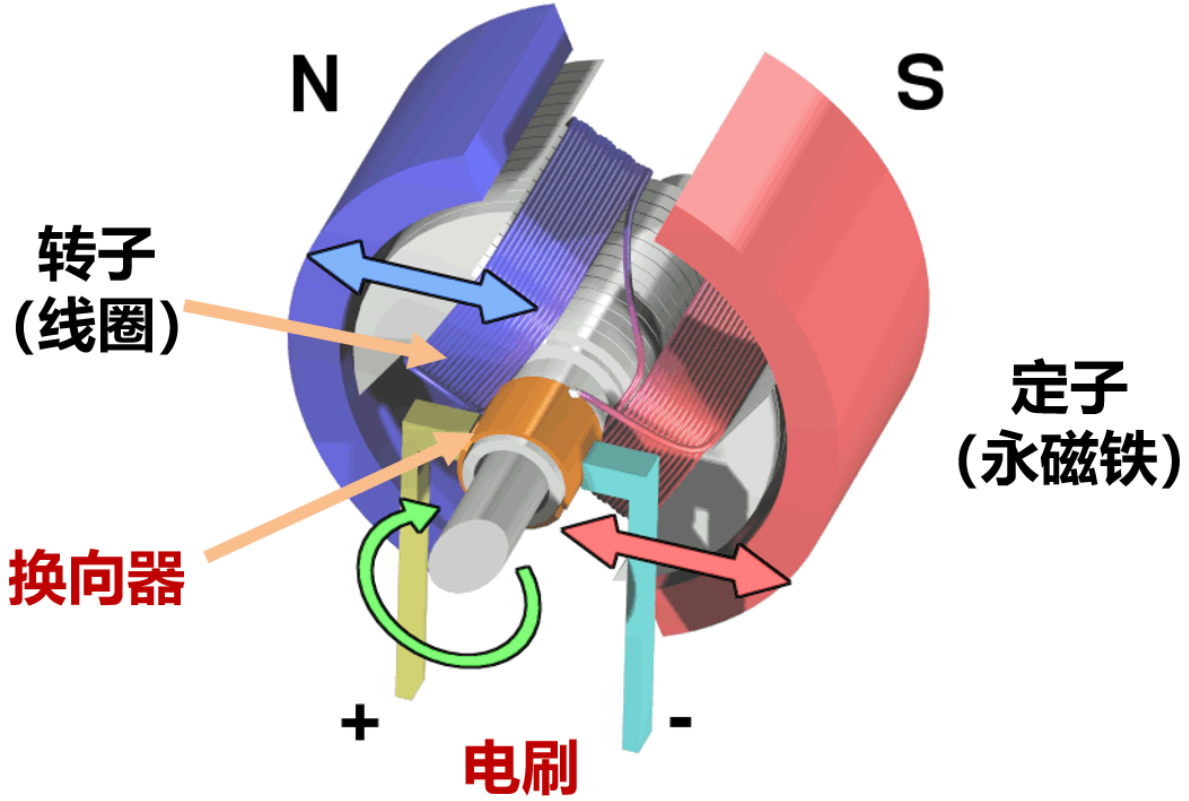
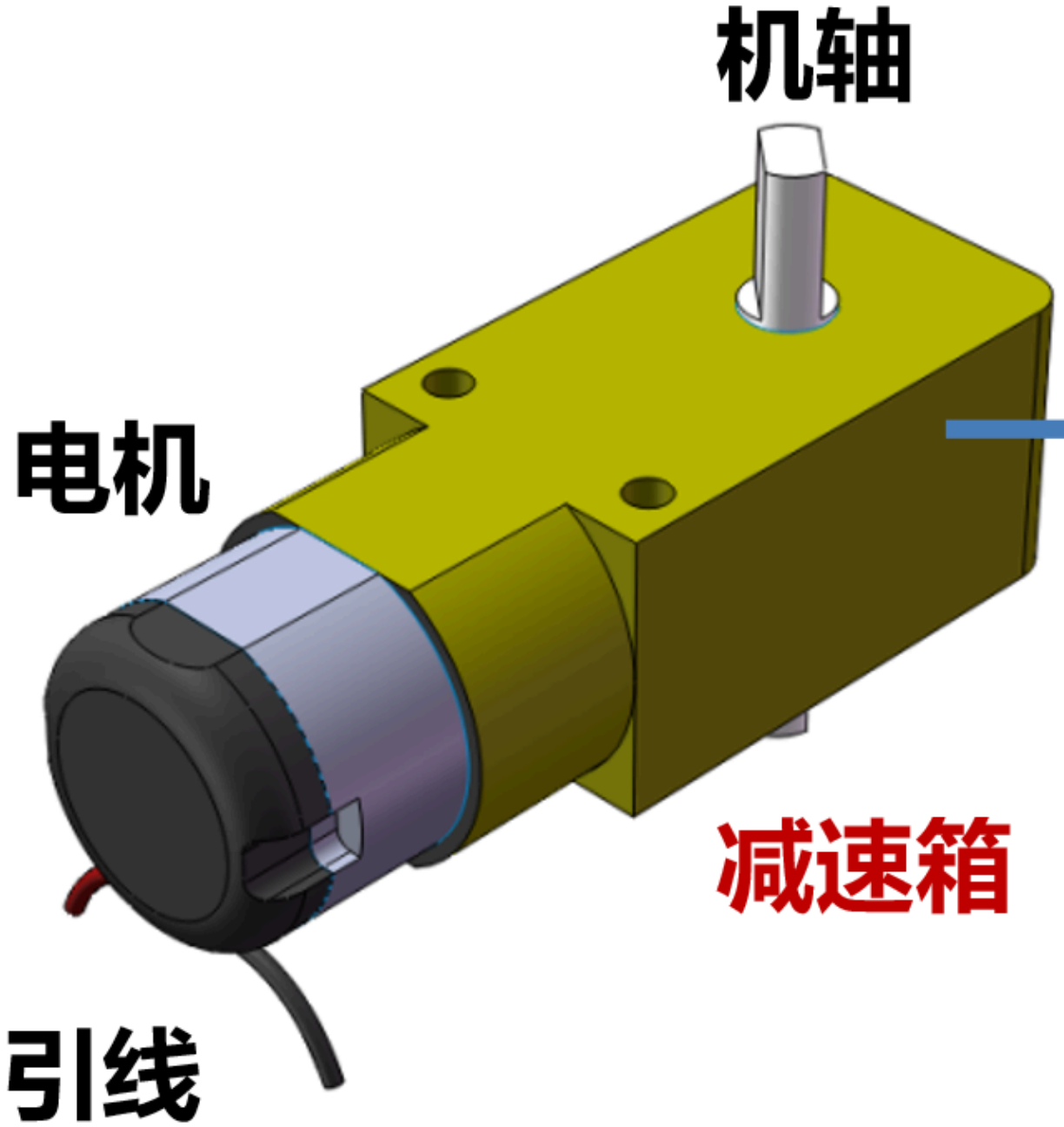
电机驱动原理

有刷直流电机

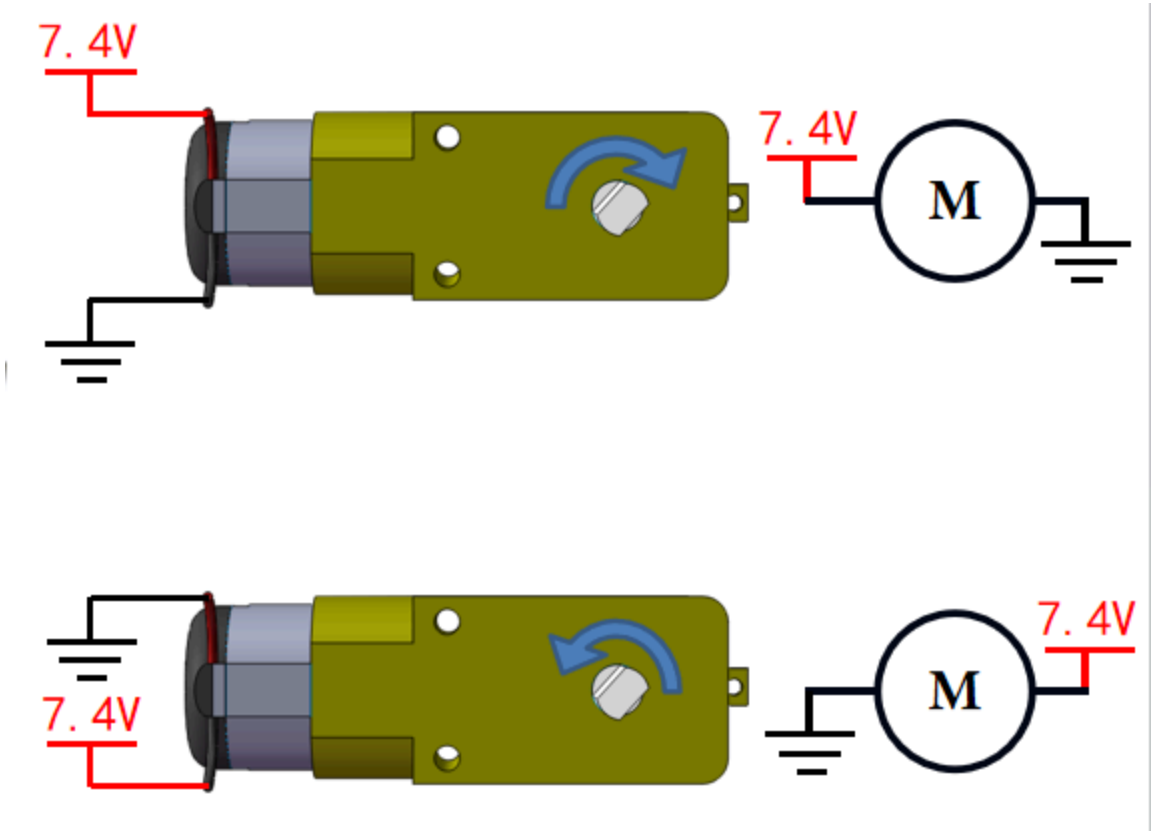
Brushed DC Motor

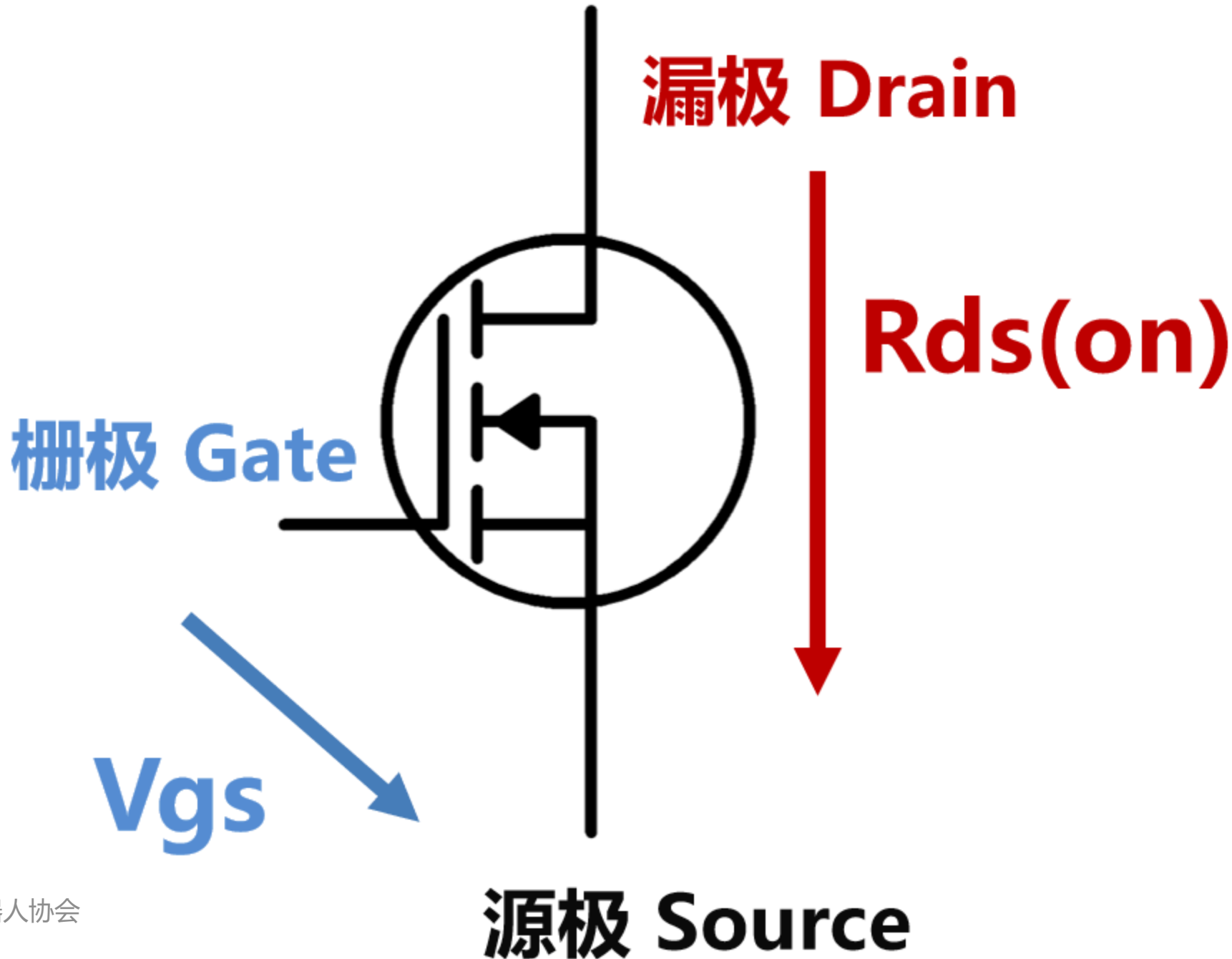


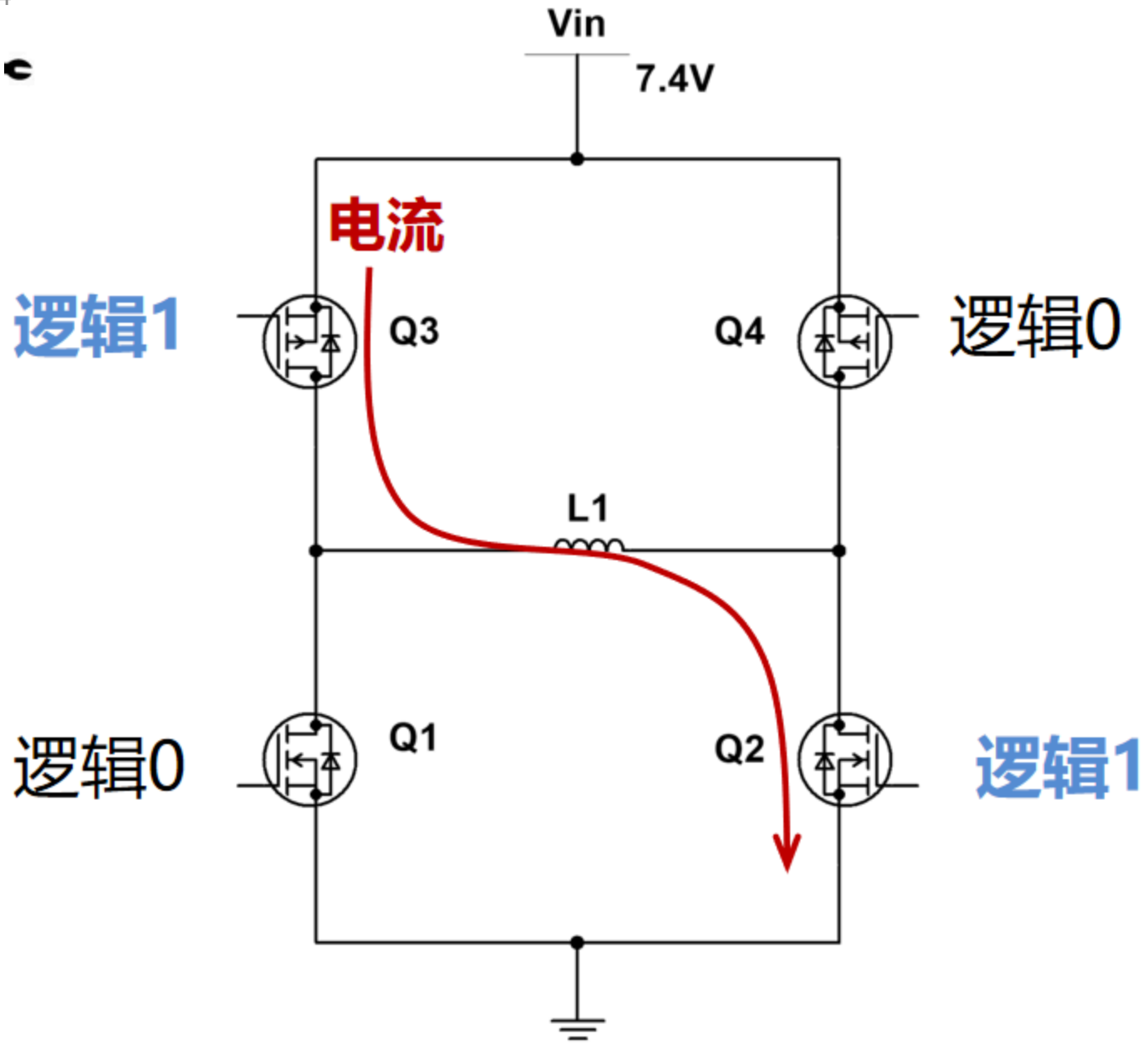


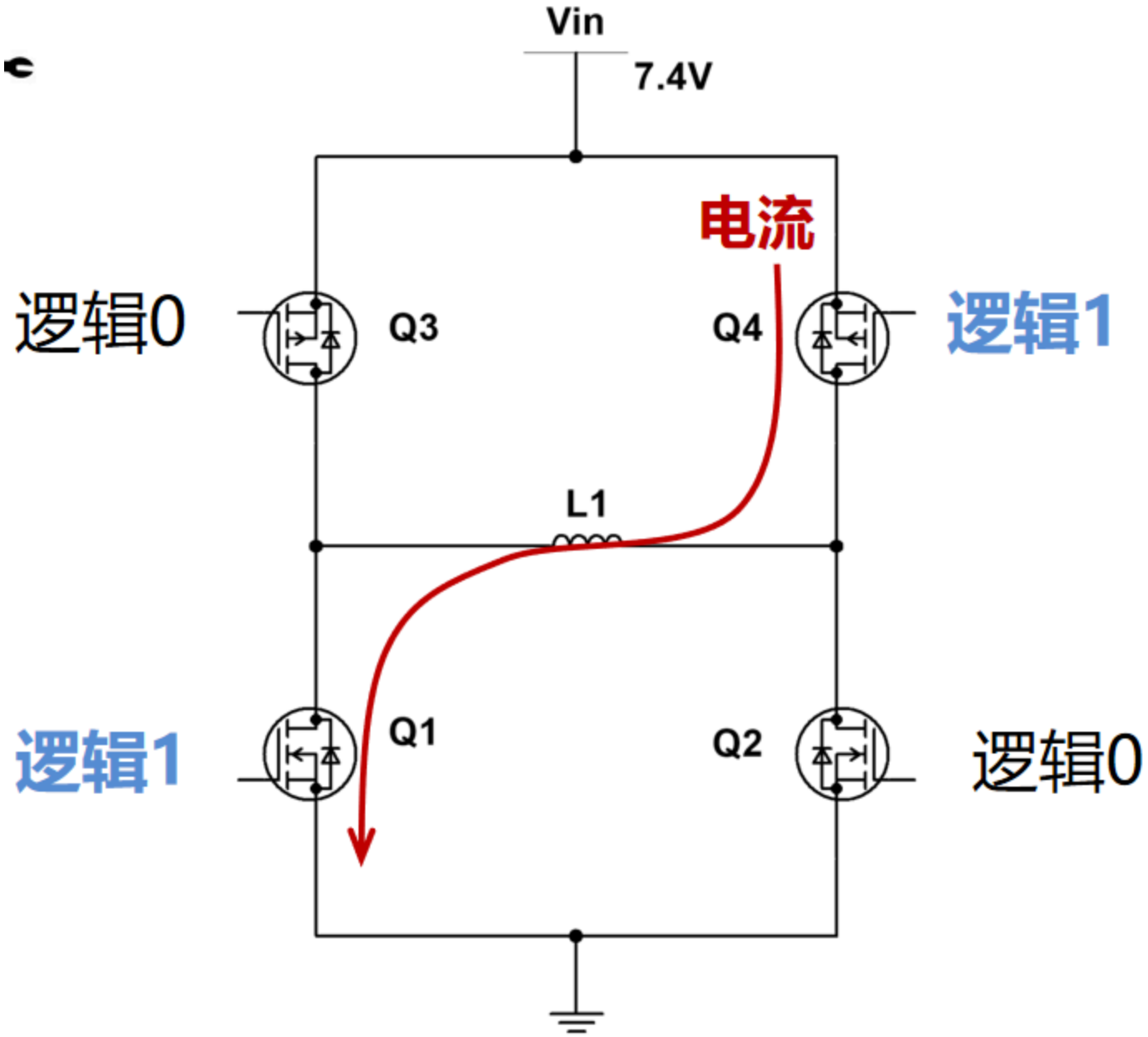
以齿轮的组合
减小转速，加大扭矩

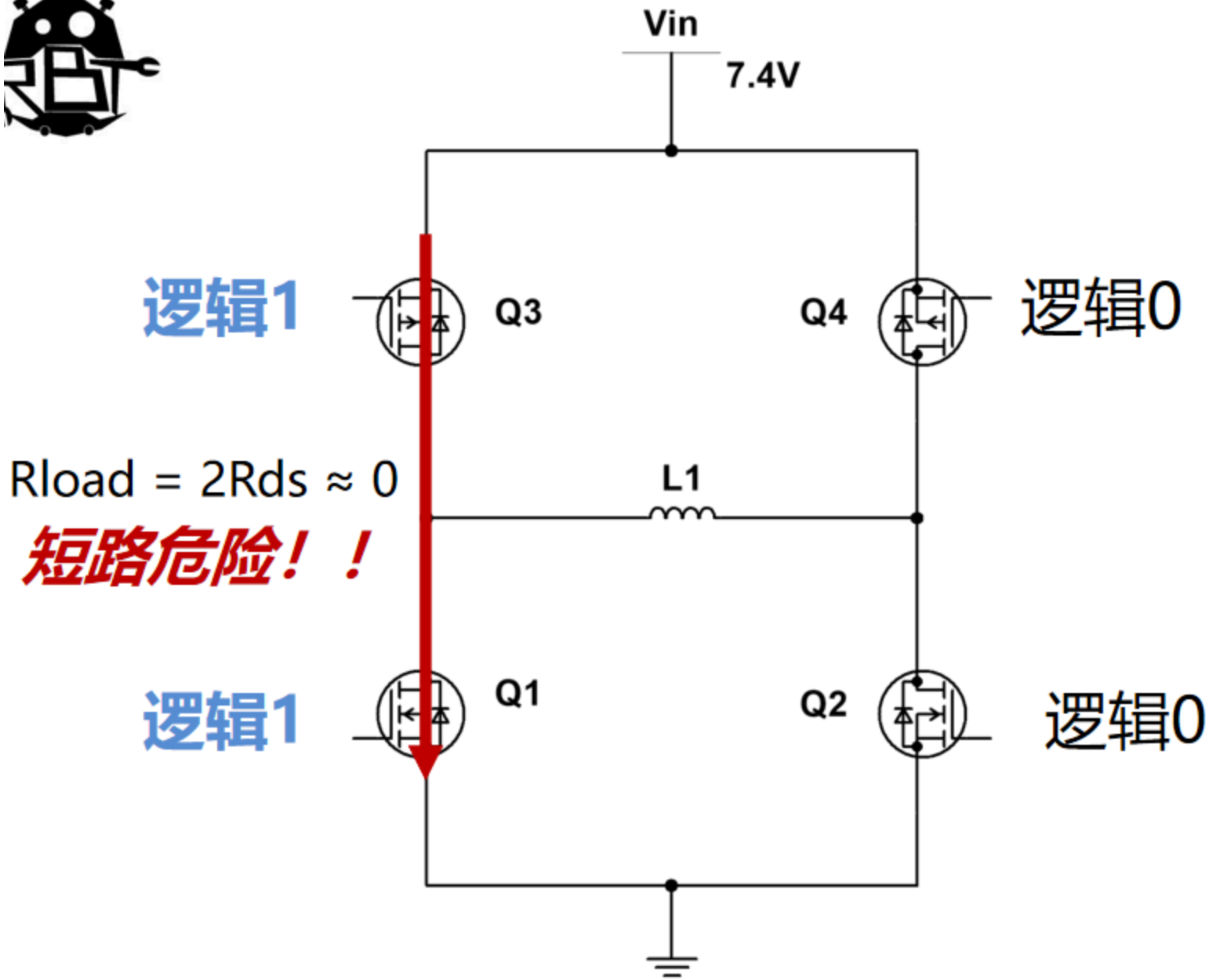
直流电机的正反转

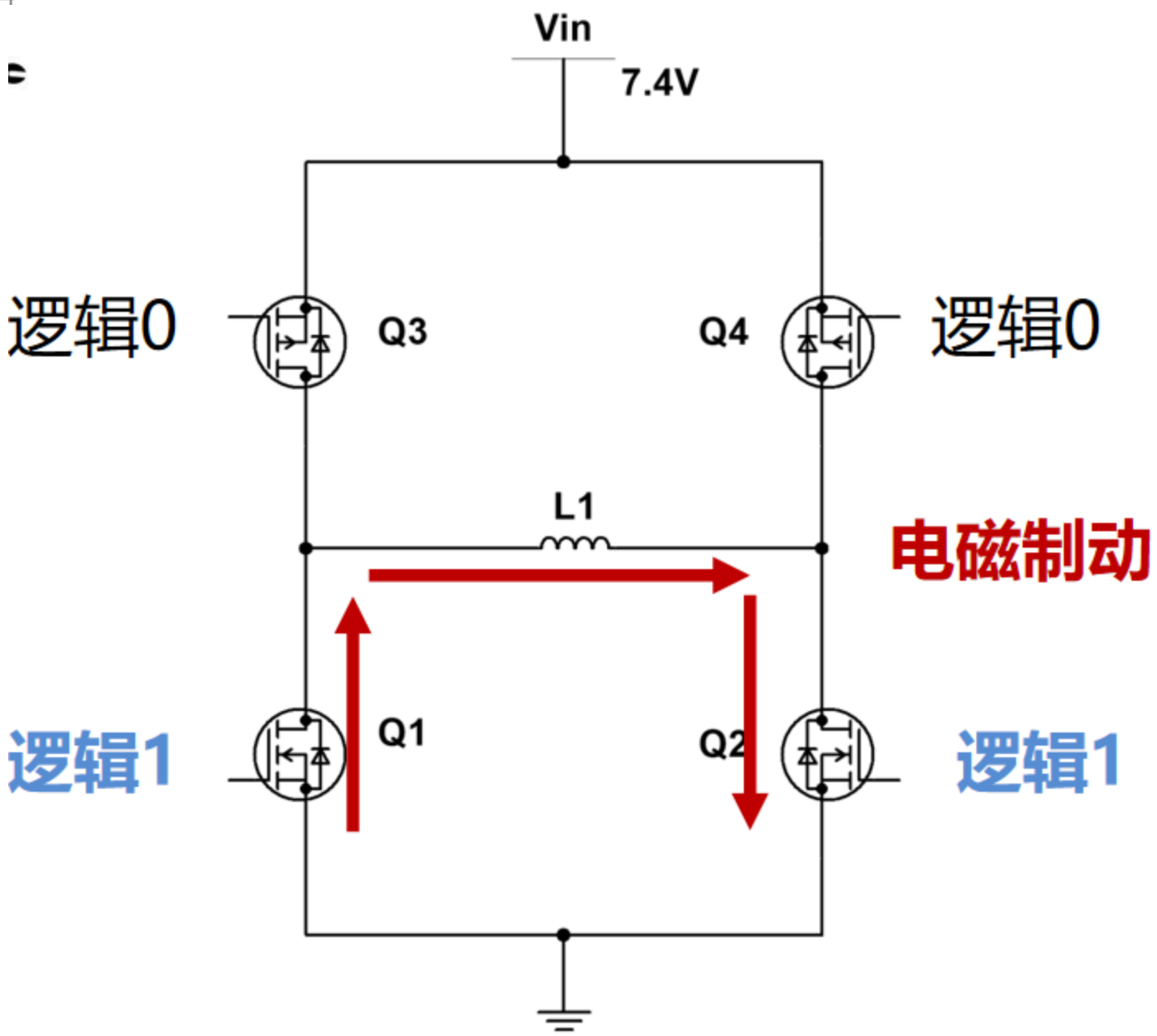


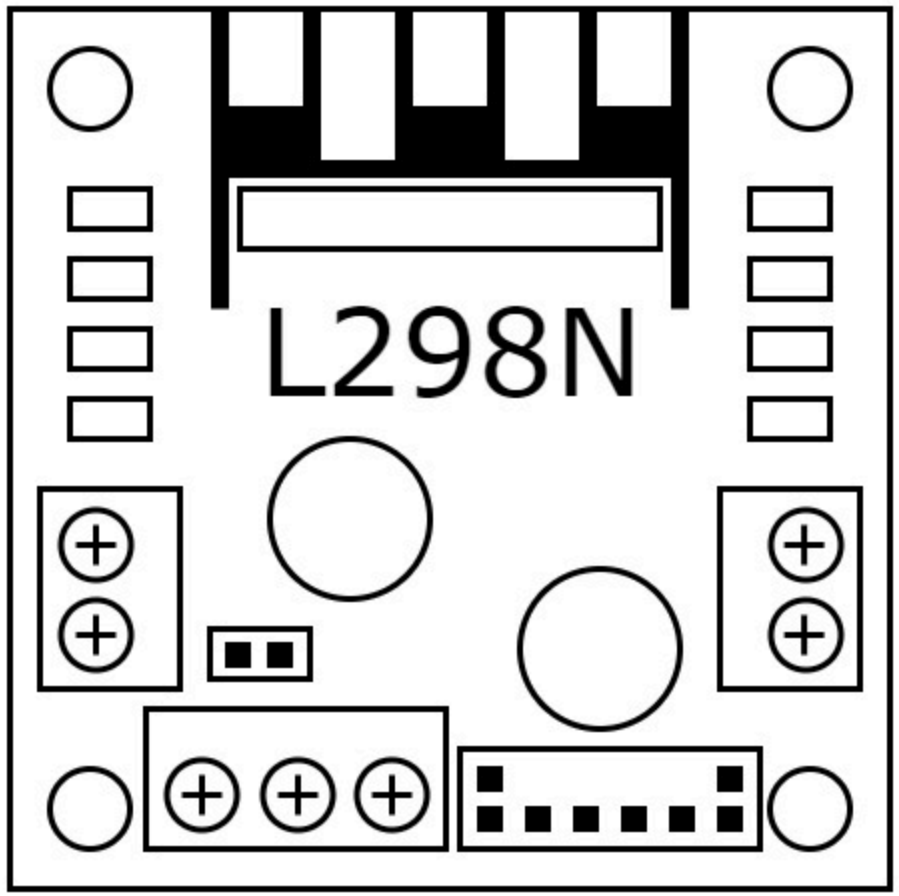

漏极 Drain
Rds(on)
栅极 Gate
Vgs
源极 Source

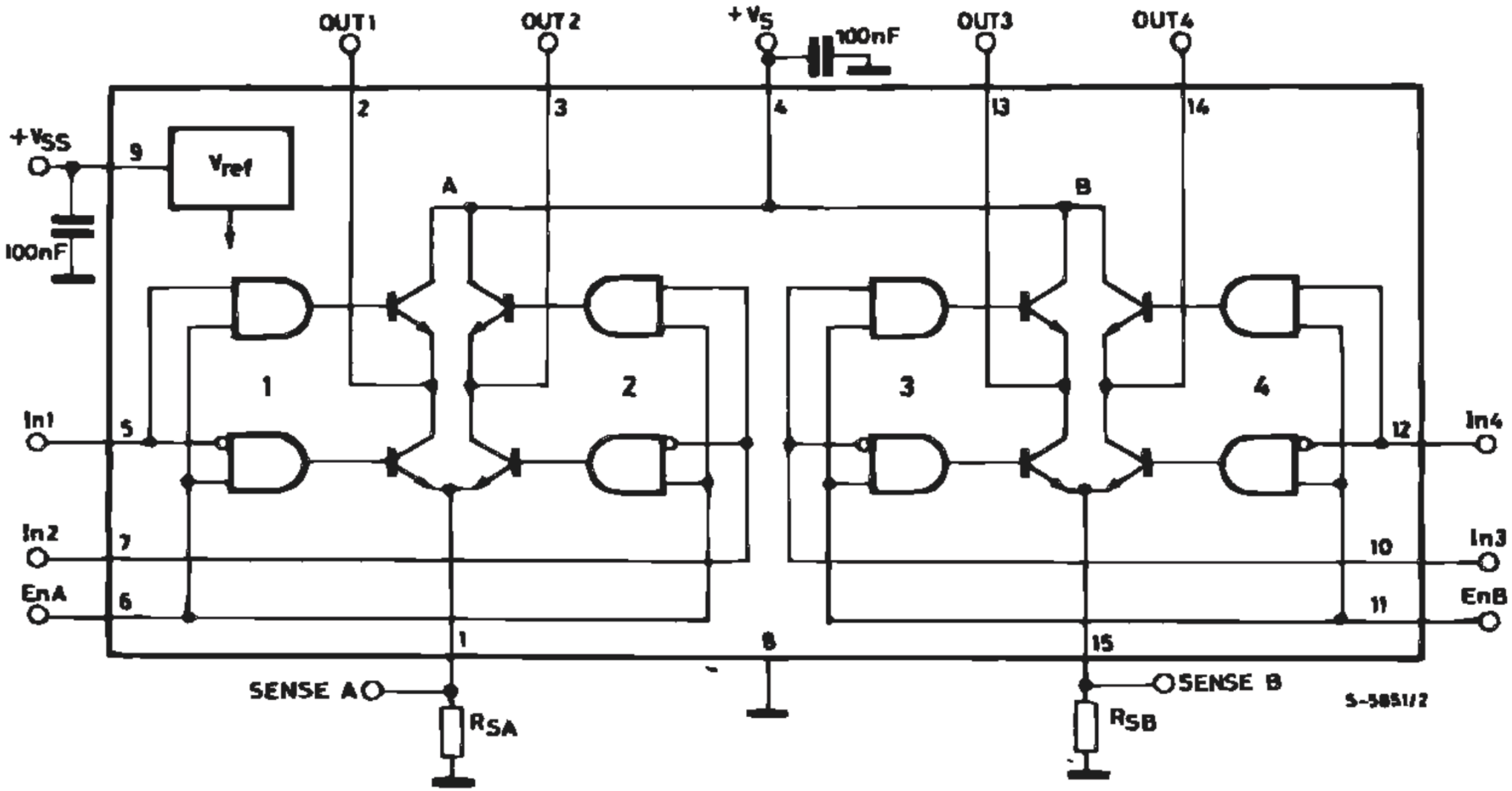

Vin
7.4V
电流
逻辑1
Q3
Q4
逻辑0
L1
逻辑0
Q1
Q2
逻辑1

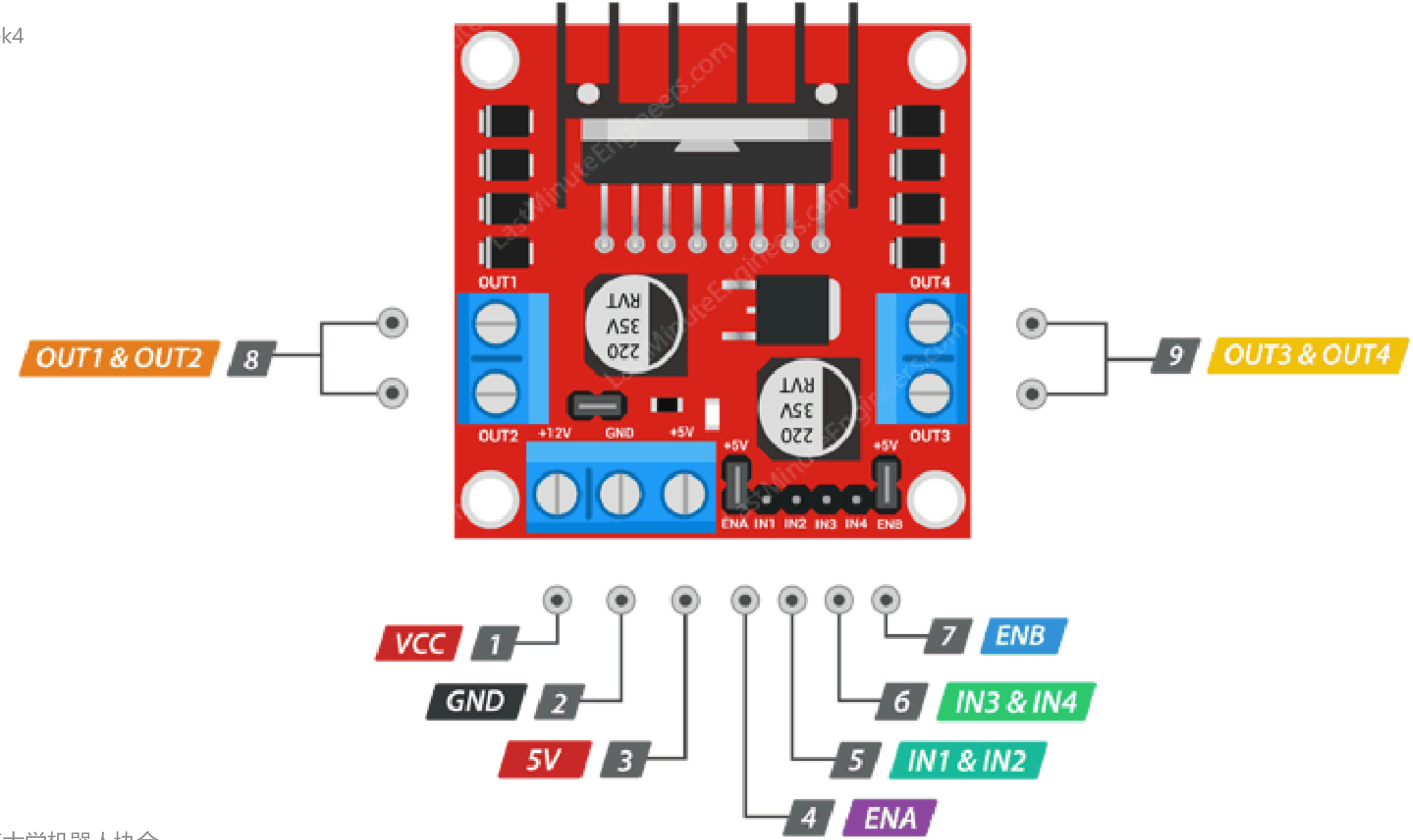

Vin
7.4V
电流
逻辑0
Q3
Q4
逻辑1
L1
逻辑1
Q1
Q2
逻辑0

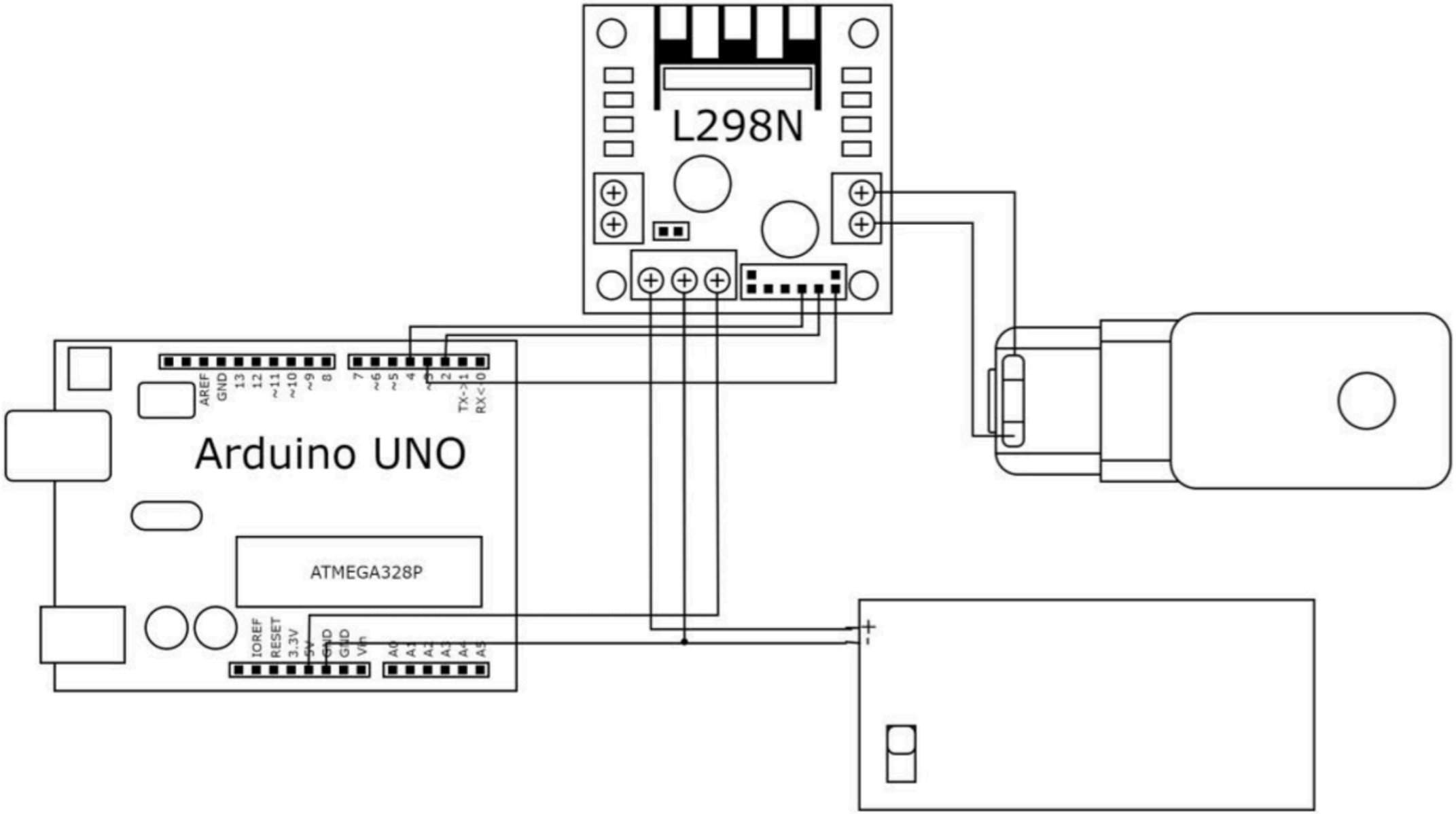

Vin
7.4V
逻辑1
Q3
Q4
逻辑0
Rload = 2Rds ≈ 0
L1
短路危险！！
逻辑1
Q1
Q2
逻辑0


Vin
7.4V
逻辑0
Q3
Q4
逻辑0
L1
电磁制动
逻辑1
Q1
Q2
逻辑1


L298N

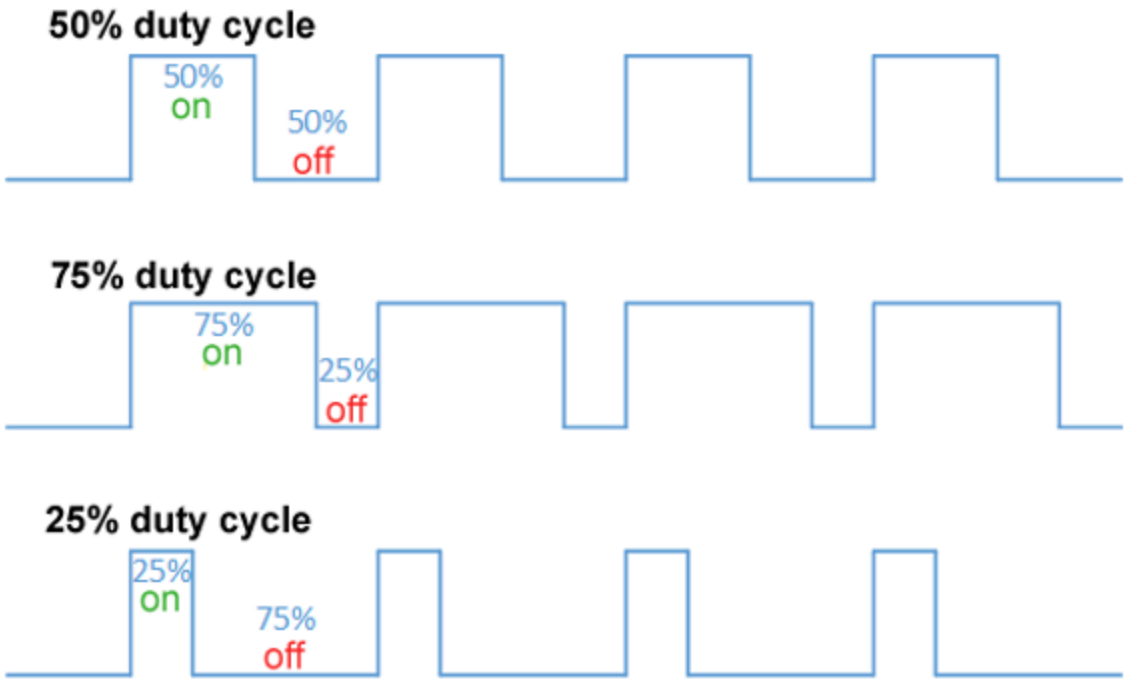

OUT1
OUT2
+VS
100nF
OUT3
OUT4
2
3
4
13
14
+VSS
9
Vref
100nF
A
B
1
2
3
4
In1
5
In2
7
EnA
6
12
In4
10
In3
11
EnB
1
8
15
SENSE A
RSA
SENSE B
RSB
S-5851/2


OUT1 & OUT2
8
OUT1
OUT2
+12V
GND
+5V
OUT4
OUT3
9
OUT3 & OUT4
ENA IN1 IN2 IN3 IN4 ENB
VCC
1
GND
2
5V
3
4
ENA
5
IN1 & IN2
6
IN3 & IN4
7
ENB


L298N
Arduino UNO
ATMEGA328P
AREF
GND
13
12
~11
~10
~9
8
7
~6
~5
4
~3
2
TX->1
RX<-0
IOREF
RESET
3.3V
5V
GND
GND
Vin
A0
A1
A2
A3
A4
A5
+
-

EnA	In1	In2	状态
0	-	-	停止
1	1	0	正转
1	0	1	反转
1	1	1	急停
1	0	0	滑行

ENA =


50% duty cycle
50%
on
50%
off
75% duty cycle
75%
on
25%
off
25% duty cycle
25%
on
75%
off

占空比太低，电机不转?

绕组电流

$$u_L(t) = L\frac{di_L}{dt}$$

$$i_L(t) = \frac{1}{L}\int_0^t u_L dt$$

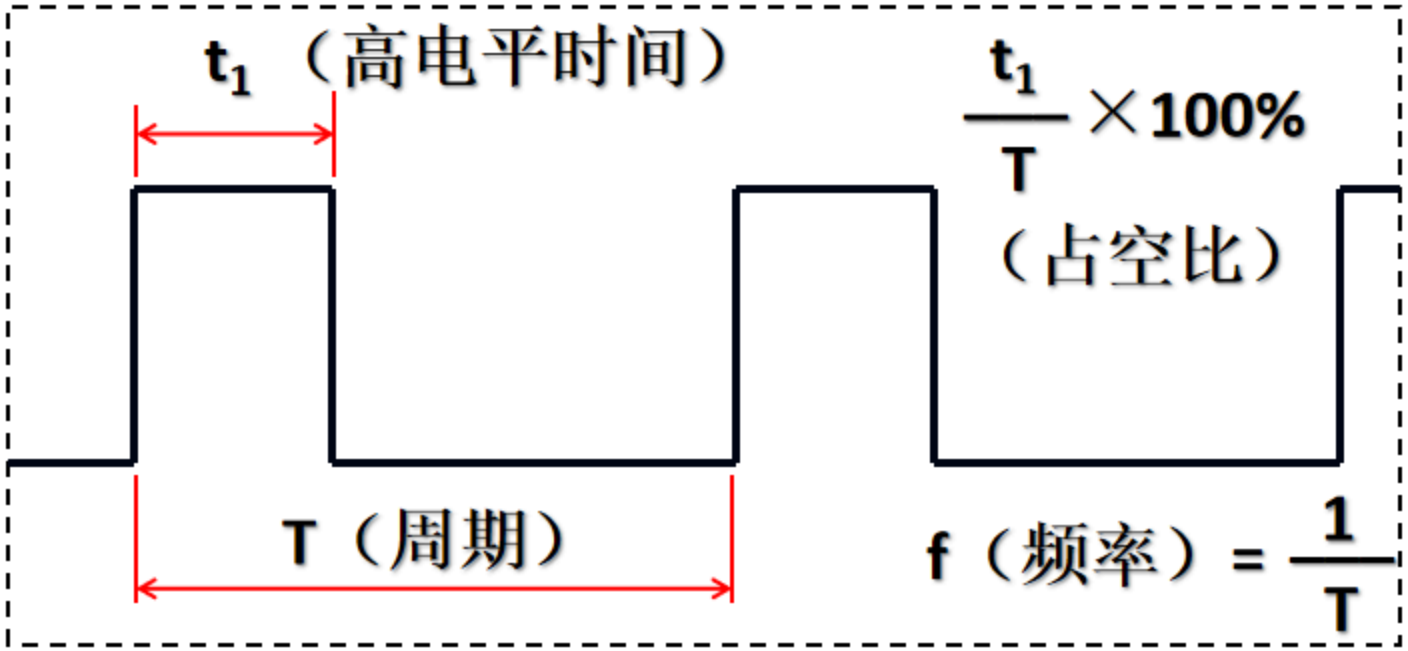
+

磁力矩

$$M = NBI\sin\theta$$

↓

绕组电感充电时间太短，电流过小
不足以克服阻力

PWM调速

- PWM（Pulse Width Modulation）脉（冲）宽（度）调制